

Line Plots

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Introduction

Line plots connect ordered observations and exploit a powerful perceptual cue: the human eye reads slope as rate of change effortlessly. They are the default geom for time-series, dose-response curves, individual-trajectory (“spaghetti”) displays, cumulative distributions, and any context where the x-axis carries a meaningful sequence. The same simplicity that makes them effective also makes them dangerous when the x-axis lacks order — connecting unordered categories with a line implies a trajectory that does not exist and misleads readers into seeing trends in noise.

Prerequisites

A working knowledge of `ggplot2`’s aesthetic mappings, a clear distinction between continuous, ordered, and unordered variables, and an understanding of how `group` aesthetics control which points get joined.

Theory

`geom_line()` joins points in the order they appear within each group defined by the `group` aesthetic; default groups come from any discrete colour, fill, or linetype mapping. For paired or repeated-measures data, the `group` aesthetic must be set explicitly to the subject identifier so that individual trajectories are connected, not all observations across subjects. Variants include:

- `geom_step()` for cumulative quantities such as ECDFs or Kaplan–Meier survival curves, where the underlying process changes only at observed events.
- `geom_path()` for trajectories ordered by row position rather than the x-aesthetic, useful for state-space plots.
- `geom_ribbon()` paired with a line for confidence or prediction bands.

For many overlapping lines, alpha-blending or faceting prevents the “spaghetti pile” that hides individual trajectories.

Assumptions

The x-axis values are ordered meaningfully — by time, dose, position, or rank. Connecting nominal categories with a line is a presentation error, not a stylistic choice.

R Implementation

```
library(ggplot2)

# Single time series
ggplot(economics, aes(date, unemploy)) +
  geom_line(colour = "#2A9D8F", linewidth = 1) +
  theme_minimal()

# Multiple groups
ggplot(airquality, aes(Day, Ozone, group = Month, colour = factor(Month))) +
```

```

geom_line(linewidth = 1) +
scale_colour_brewer(palette = "Set2") +
theme_minimal()

# Paired trajectories (pre/post style)
df <- data.frame(
  subject = rep(1:20, each = 2),
  time    = rep(c("pre", "post"), 20),
  value   = c(rnorm(20, 50, 10), rnorm(20, 55, 10))
)
ggplot(df, aes(time, value, group = subject)) +
  geom_line(colour = "grey70") +
  geom_point(size = 2) +
  theme_minimal()

# Step line (cumulative)
ggplot(mtcars, aes(wt)) +
  stat_ecdf(geom = "step", colour = "#2A9D8F", linewidth = 1) +
  theme_minimal()

```

Output & Results

Four illustrative line plots: a long-running unemployment series, monthly ozone levels coloured by month, individual subject trajectories from pre to post, and the empirical cumulative distribution function of car weights as a step line. Each shows a different idiomatic use of the line geom.

Interpretation

A reporting sentence: “Individual subject responses from pre to post are connected by grey lines, with overlaid points indicating measurement times; group-mean changes (not shown) are reported in Table 1.” Always describe the unit of analysis a line represents (subject, country, replicate); a line plot without that context is ambiguous.

Practical Tips

- Set the `group` aesthetic explicitly for paired or repeated-measures data; the default group inferred from colour or linetype rarely matches the intended unit.
- Add `geom_point()` on top of `geom_line()` when x-values are sparse so individual observations remain visible; pure lines hide irregular sampling.
- For uncertainty bands, layer `geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.3)` *before* the line so the line draws on top.
- Use `geom_step()` for ECDFs and survival curves — they are step functions, not smooth ones, and a smooth interpolation misrepresents the underlying mathematics.
- For dozens of overlapping trajectories, lower `alpha` (0.2–0.4) and add a single bold line for the group mean or median; pure spaghetti loses readability above 30–40 trajectories.
- When the y-axis spans orders of magnitude with multiplicative growth, use `scale_y_log10()` so constant-rate growth reads as a straight line.

R Packages Used

`ggplot2` for `geom_line()`, `geom_step()`, `geom_path()`, `geom_ribbon()`, and `stat_ecdf()`; `ggrepel` for non-overlapping line-end labels; `directlabels` for legend-free per-series labelling that scales better than colour-and-legend in print.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts